

surveydown

An Open-Source, Markdown-Based Platform
for Reproducible and Programmable Surveys

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About Me



- **Pingfan Hu**, PhD Candidate at George Washington University, advised by Dr. John Helveston.
- Research interests:
 1. **Data Science** in sustainable transportation and human behavior
 2. **Open-Source Software** in **R** tools for survey research
 3. **Agentic AI Engineering** around Claude Code
- For more information, visit pingfanhu.com

I try to figure out what people want...





Prof. John
Helveston



jhelvy.shinyapps.io



Welcome to the Ultimate Superhero Survey!



Thank you for participating in our research on superhero preferences and powers! This survey will help us understand what makes superheroes appealing to different people.

This survey should take about 3-5 minutes to complete.

Do you have a favorite super hero?

*

☐ Yes

☒ No

Next

WYSIWYG Interface

The screenshot shows a Google Docs document titled "Super Hero Survey!". The interface includes a top navigation bar with "Questions", "Responses", and "Settings" tabs. A dropdown menu is open, displaying various question types: Short answer, Paragraph, Multiple choice (selected), Checkboxes, Dropdown, File upload, Linear scale, Rating (marked as "New"), Multiple choice grid, Checkbox grid, Date, and Time. The main content area shows a survey question: "Do you have a favorite super hero?" with radio button options for "Yes", "No", and "Add option or add 'Other'". A "Publish" button is visible in the top right corner.

docs.google.com

Super Hero Survey! All changes saved in Drive

Questions Responses Settings

Super Hero Survey!

Form description

Do you have a favorite super hero?

☐ Yes

☐ No

☐ Add option or [add "Other"](#)

Short answer

Paragraph

☒ Multiple choice

☒ Checkboxes

☐ Dropdown

☒ File upload

Linear scale

★ Rating **New**

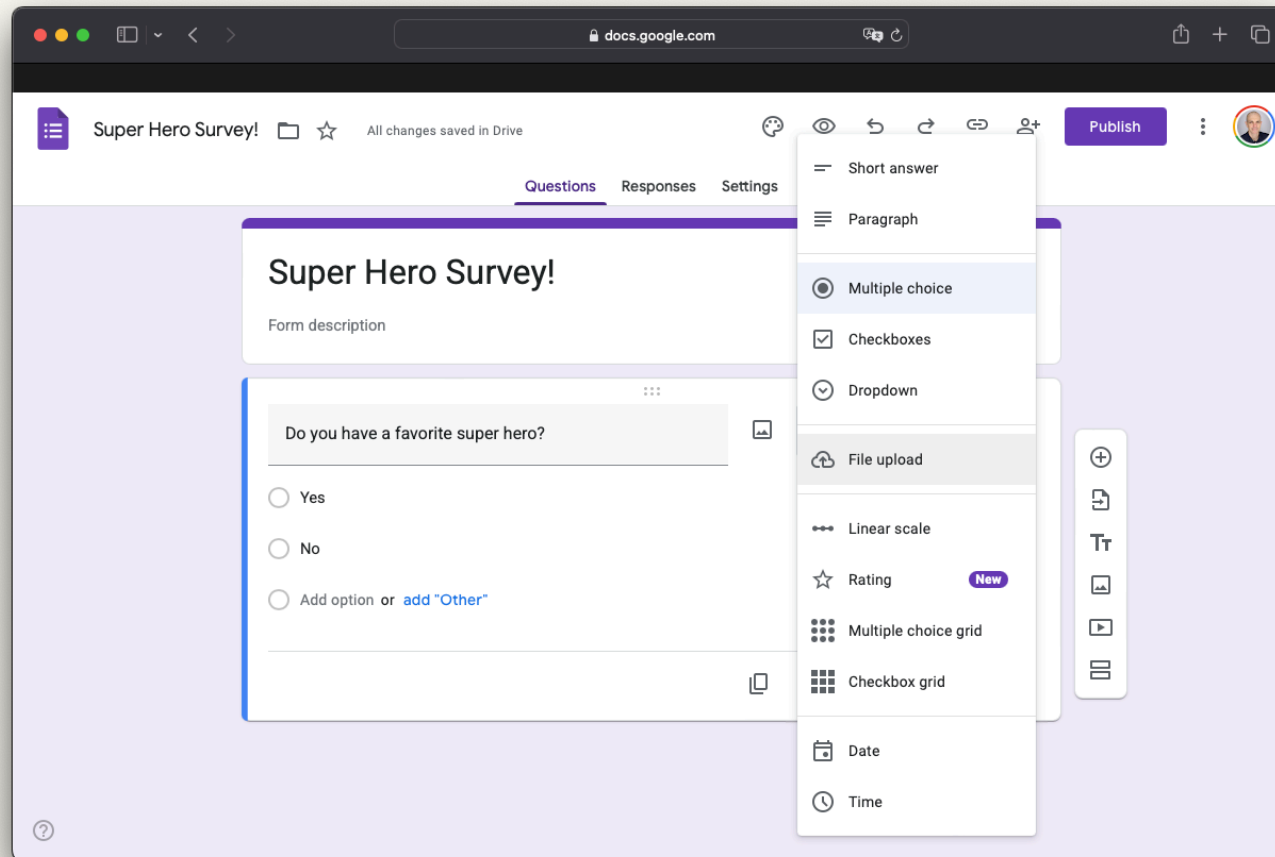
☒ Multiple choice grid

☒ Checkbox grid

☐ Date

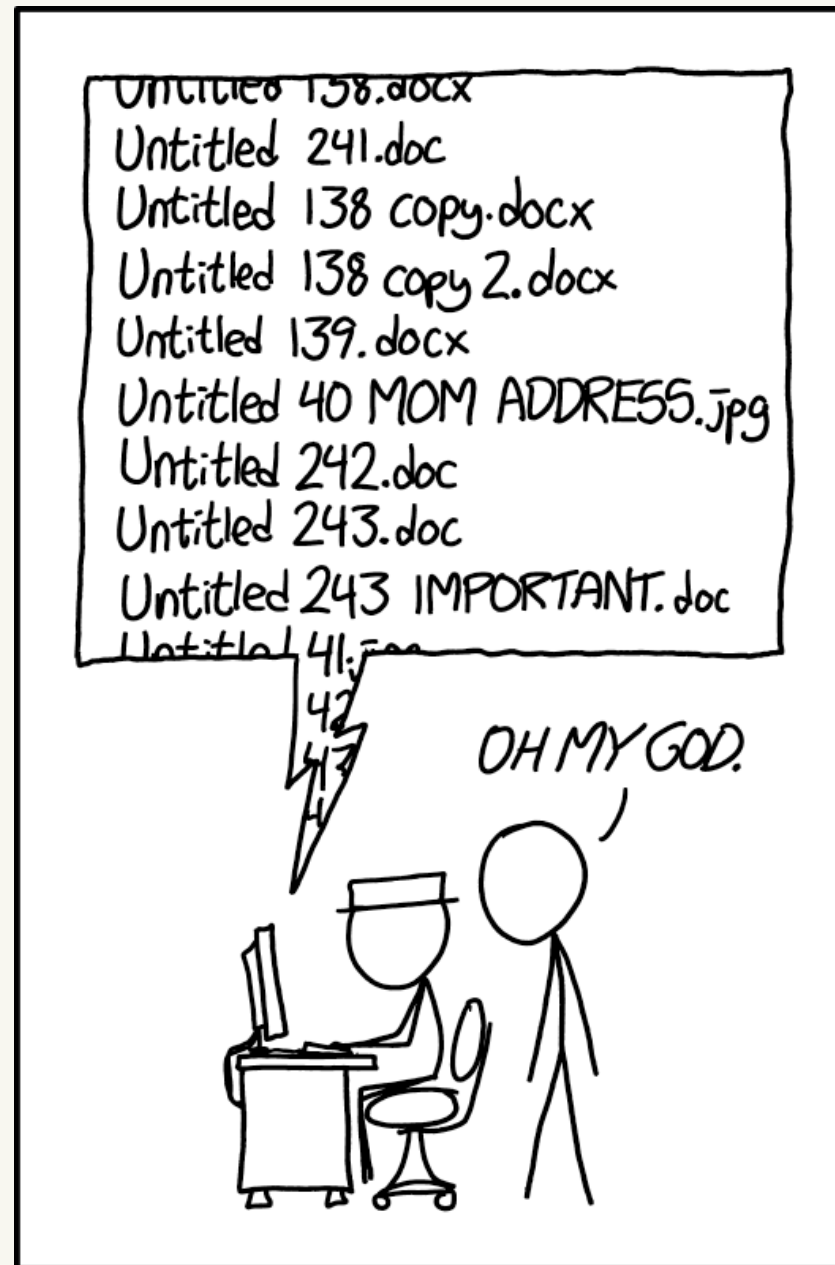
☐ Time

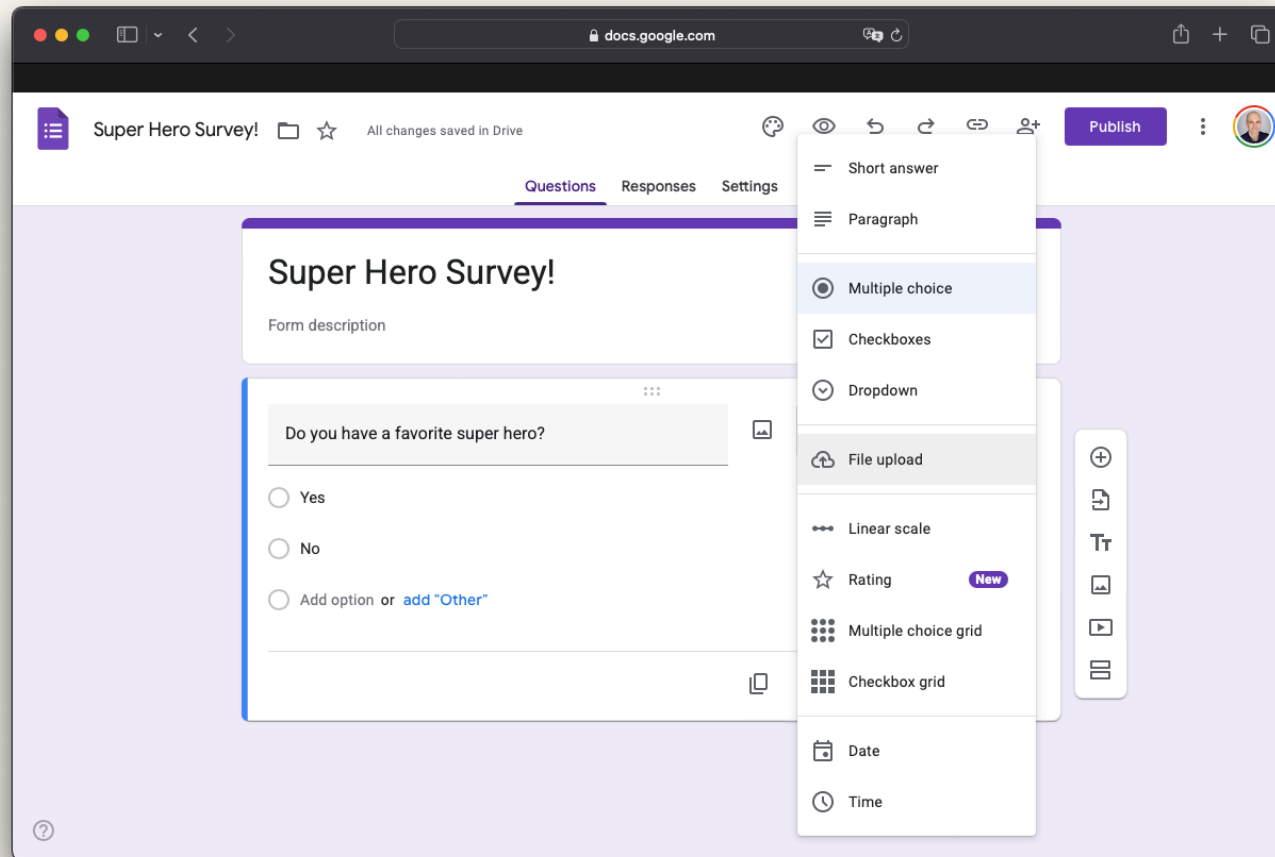
Publish



Limitations

- ✗ Reproducibility
- ✗ Version control





Limitations

- ✗ Reproducibility
- ✗ Version control
- ✗ Limited features
- ✗ Open source

Why not make surveys
from **code**?

~~Limitations~~

Features

- ✓ Reproducibility
- ✓ Version control
- ✓ Lots of features
- ✓ Open source

Introducing **surveydown**!



In this talk, we will cover:

What is surveydown?

How does it work?

What can I do with it?

What's next?

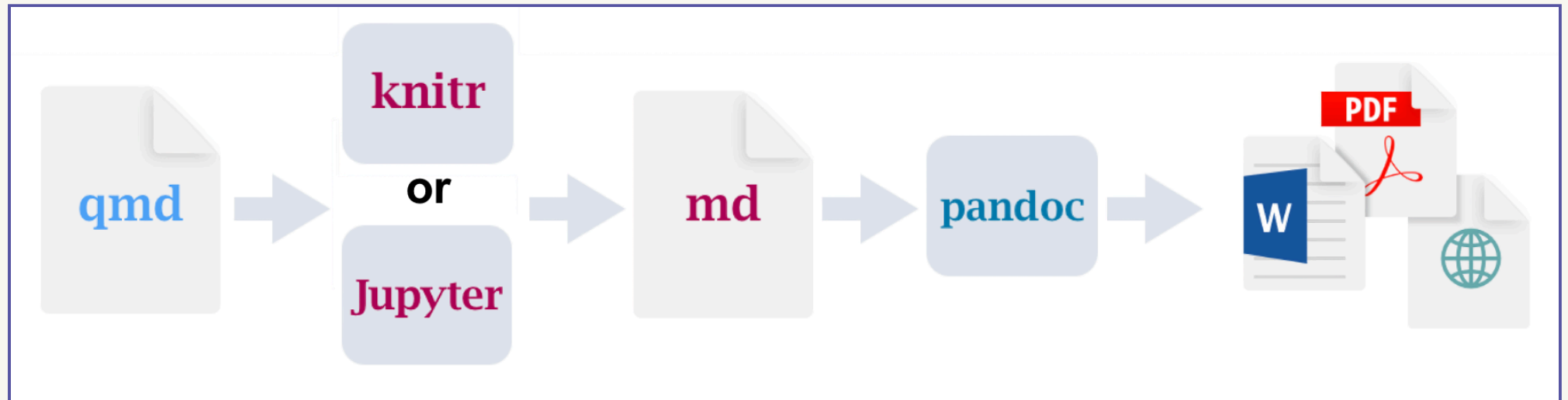
What is **surveydown**?



R package that renders Quarto files into surveys



Quarto is a publishing system

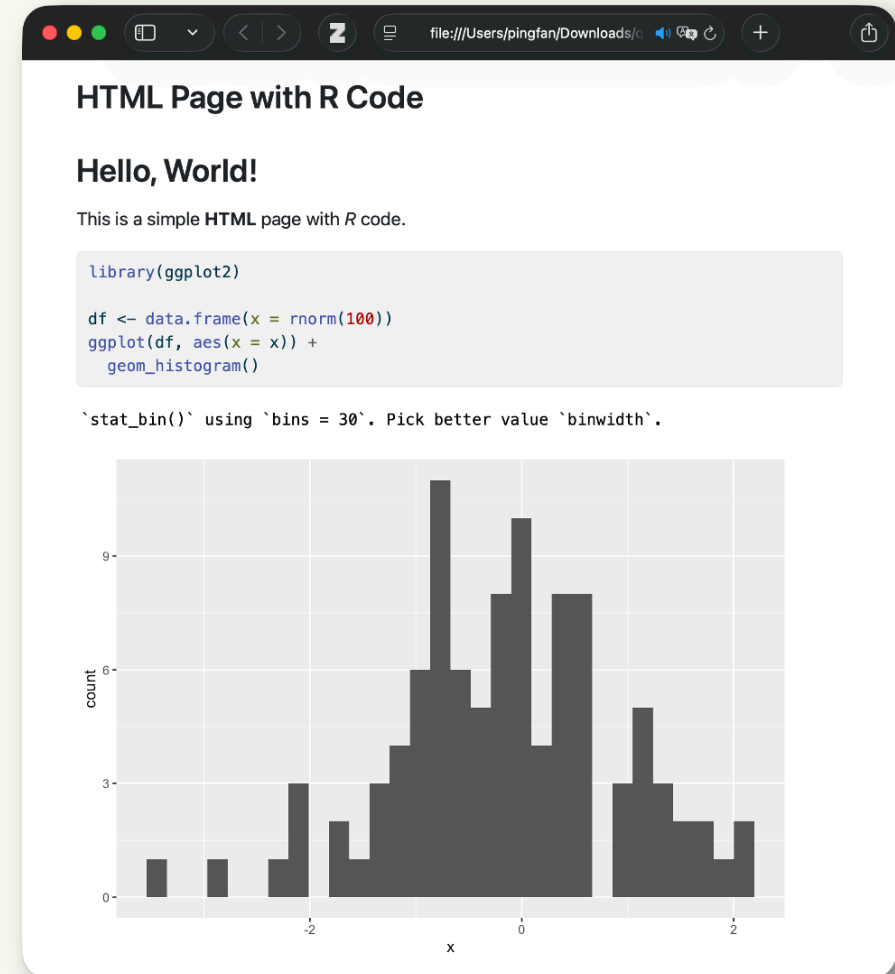


Original qmd file

Markdown + R code chunks

```
1 ---
2 format: html
3 title: "HTML Page with R Code"
4 ---
5
6 # Hello, World!
7
8 This is a simple **HTML** page with *R* code.
9
10 ```{r}
11 library(ggplot2)
12
13 df <- data.frame(x = rnorm(100))
14 ggplot(df, aes(x = x)) +
15   geom_histogram()
16 ```
```

Rendered HTML

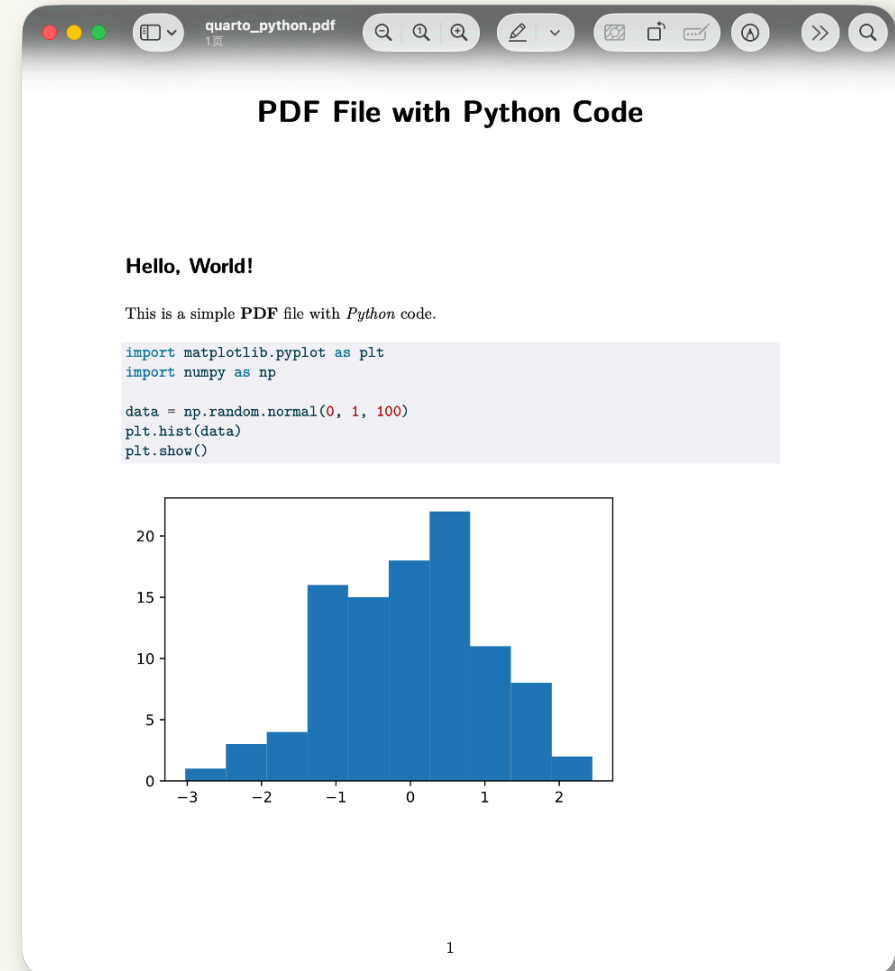


Original qmd file

Markdown + Python code chunks

```
1 ---
2 format: pdf
3 title: "PDF File with Python Code"
4 ---
5
6 # Hello, World!
7
8 This is a simple **PDF** file with *Python* code.
9
10 ```{python}
11 import matplotlib.pyplot as plt
12 import numpy as np
13
14 data = np.random.normal(0, 1, 100)
15 plt.hist(data)
16 plt.show()
17 ```
```

Rendered PDF

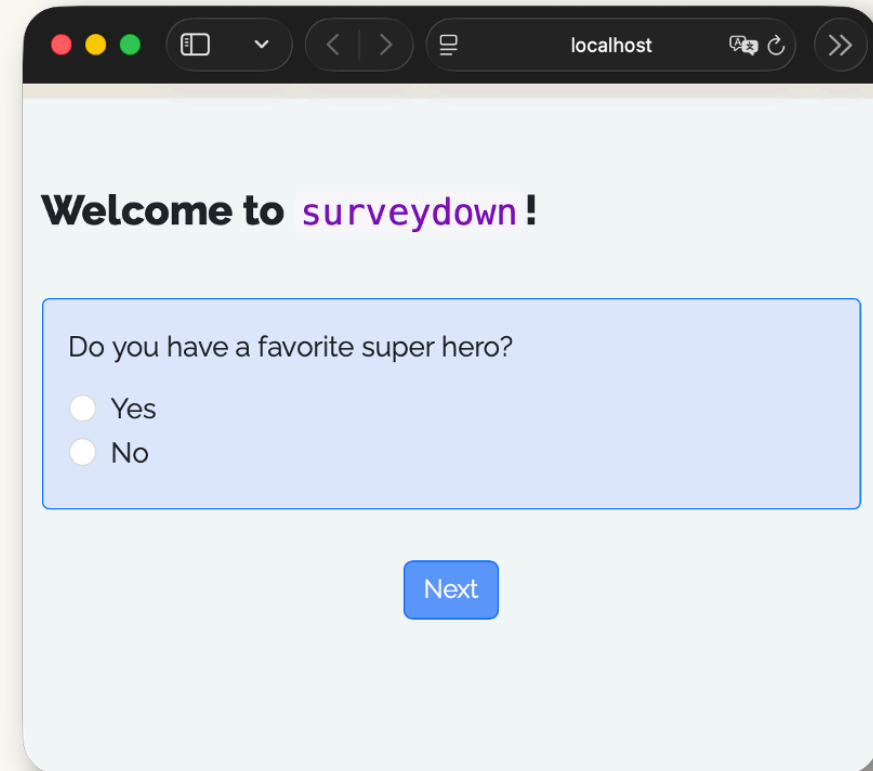




survey.qmd

```
1 ---
2 format: html
3 echo: false
4 warning: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to `surveydown`!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id    = "has_fav_hero",
19   label = "Do you have a favorite superhero?",
20   option = c(
21     "Yes" = "yes",
22     "No"  = "no"
23   )
24 )
25
26 sd_next()
27 ```
28
29 --- some_other_page
30
31 Other content...
```

Rendered Survey



localhost

Welcome to **surveydown**!

Do you have a favorite super hero?

☐ Yes

☐ No

Next

survey.qmd

YAML header for a “clean” output

```
1 ---
2 format: html
3 echo: false
4 warning: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to `surveydown`!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id    = "has_fav_hero",
19   label = "Do you have a favorite superhero?",
20   option = c(
21     "Yes" = "yes",
22     "No"  = "no"
23   )
24 )
25
26 sd_next()
27 ```
28
29 --- some_other_page
30
31 Other content...
```

survey.qmd

```
1 ---
2 format: html
3 echo: false
4 warning: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to `surveydown`!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id   = "has_fav_hero",
19   label = "Do you have a favorite superhero?",
20   option = c(
21     "Yes" = "yes",
22     "No"  = "no"
23   )
24 )
25
26 sd_next()
27 ```
28
29 --- some_other_page
30
31 Other content...
```

Load the **surveydown** Package

survey.qmd

```
1 ---
2 format: html
3 echo: false
4 warning: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to `surveydown`!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id = "has_fav_hero",
19   label = "Do you have a favorite superhero?",
20   option = c(
21     "Yes" = "yes",
22     "No" = "no"
23   )
24 )
25
26 sd_next()
27 ```
28
29 --- some_other_page
30
31 Other content...
```

Use triple-dash marker (---) to define survey pages

Quarto fences (:::) also OK:

```
::: {#welcome .sd-page}
```

Page content

```
:::
```

survey.qmd

```
1 ---
2 format: html
3 echo: false
4 warning: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to `surveydown`!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id    = "has_fav_hero",
19   label = "Do you have a favorite superhero?",
20   option = c(
21     "Yes" = "yes",
22     "No"  = "no"
23   )
24 )
25
26 sd_next()
27 ```
28
29 --- some_other_page
30
31 Other content...
```

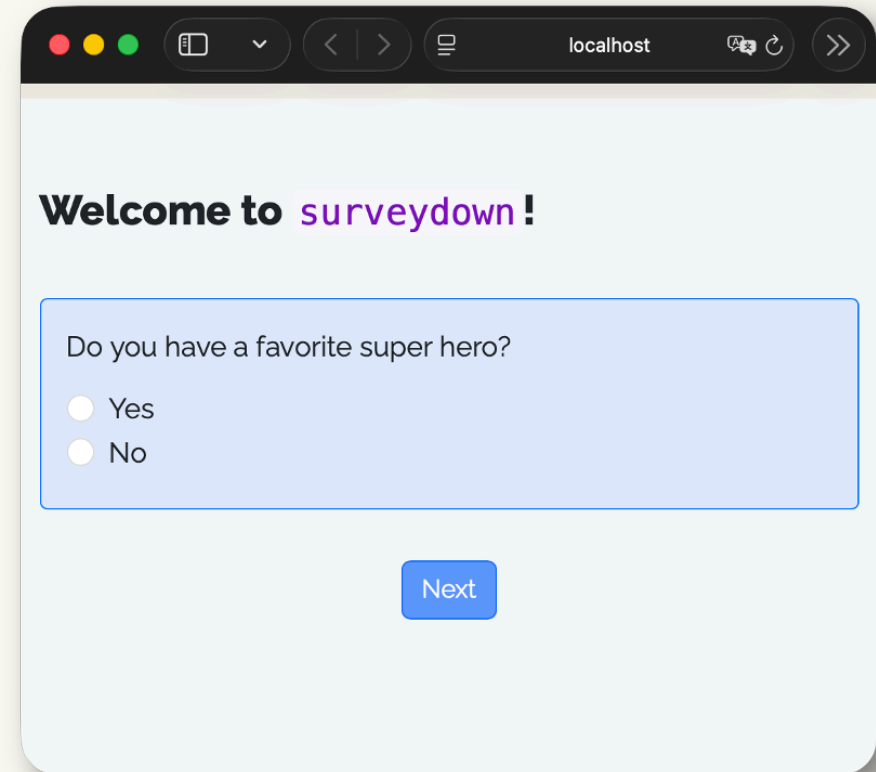
Page content

- Markdown for text, images, etc.
- `sd_question()` for survey questions
- `sd_next()` for page navigation

survey.qmd

```
1 ---
2 format: html
3 echo: false
4 warning: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to `surveydown`!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id    = "has_fav_hero",
19   label = "Do you have a favorite superhero?",
20   option = c(
21     "Yes" = "yes",
22     "No"  = "no"
23   )
24 )
25
26 sd_next()
27 ```
28
29 --- some_other_page
30
31 Other content...
```

Rendered Survey



localhost

Welcome to **surveydown**!

Do you have a favorite super hero?

☐ Yes

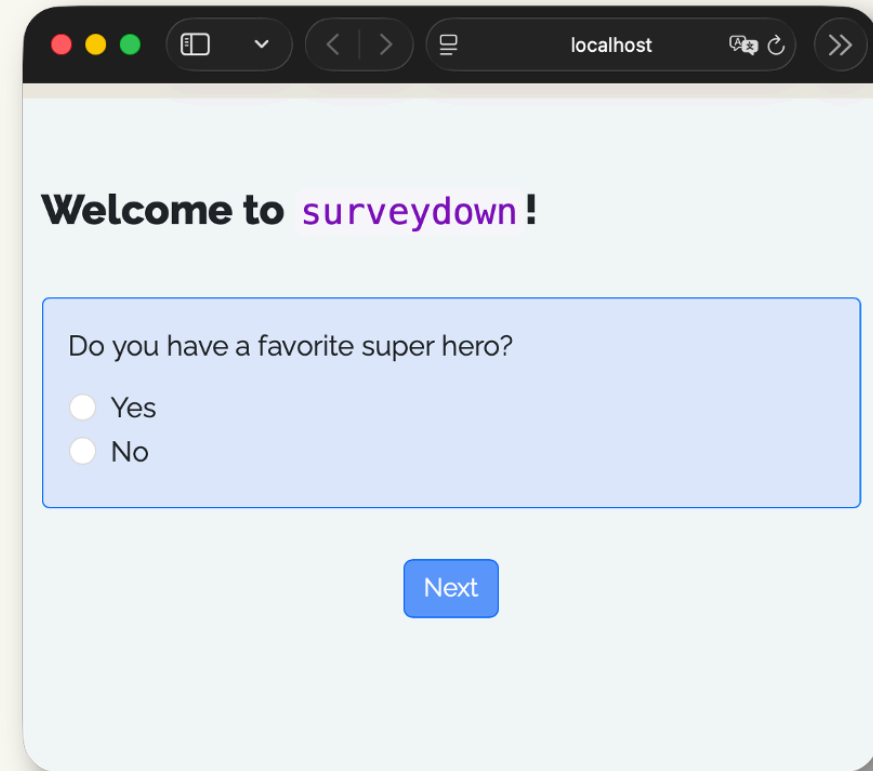
☐ No

Next

survey.qmd

```
1 ---
2 format: html
3 echo: false
4 warning: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to `surveydown`!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id    = "has_fav_hero",
19   label = "Do you have a favorite superhero?",
20   option = c(
21     "Yes" = "yes",
22     "No"  = "no"
23   )
24 )
25
26 sd_next()
27 ```
28
29 --- some_other_page
30
31 Other content...
```

Rendered Survey



Welcome to surveydown!

Do you have a favorite super hero?

☐ Yes

☐ No

Next

has_fav_hero
yes
no
no
yes

Wait a minute...

Quarto renders to **static** html pages, right?



Shiny to the rescue!





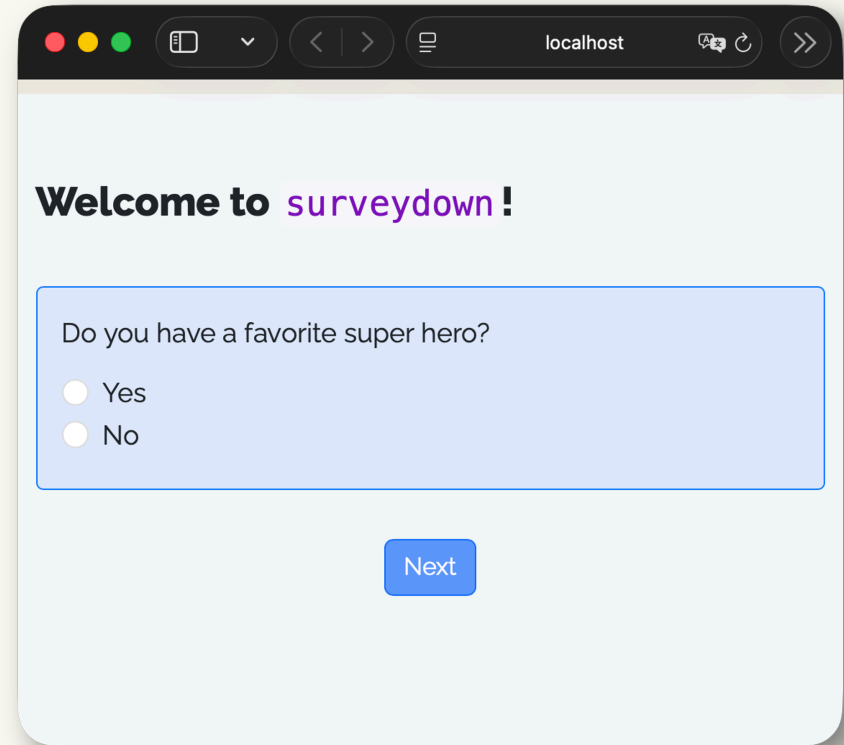
Shiny App

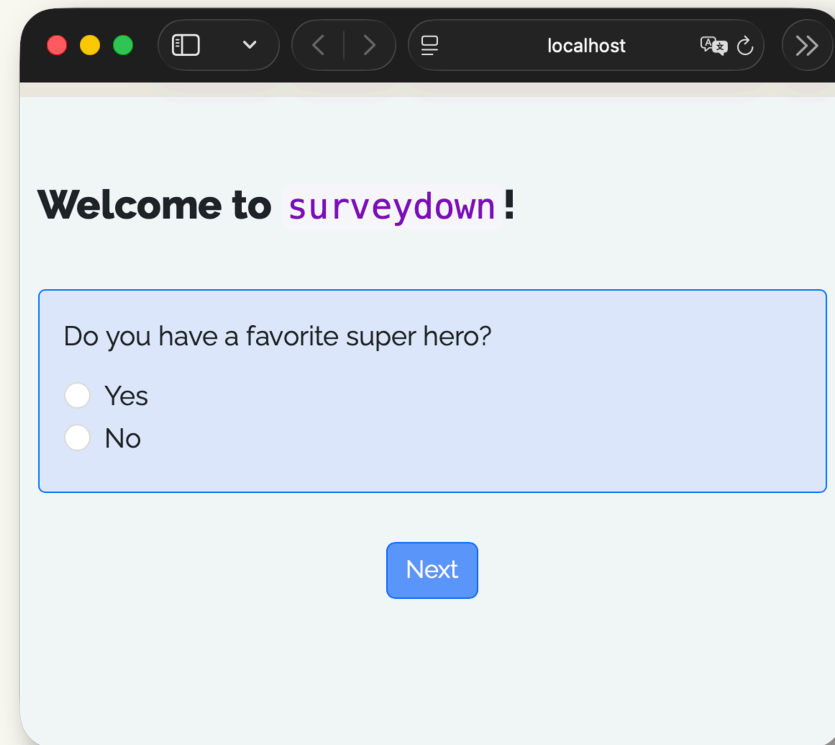
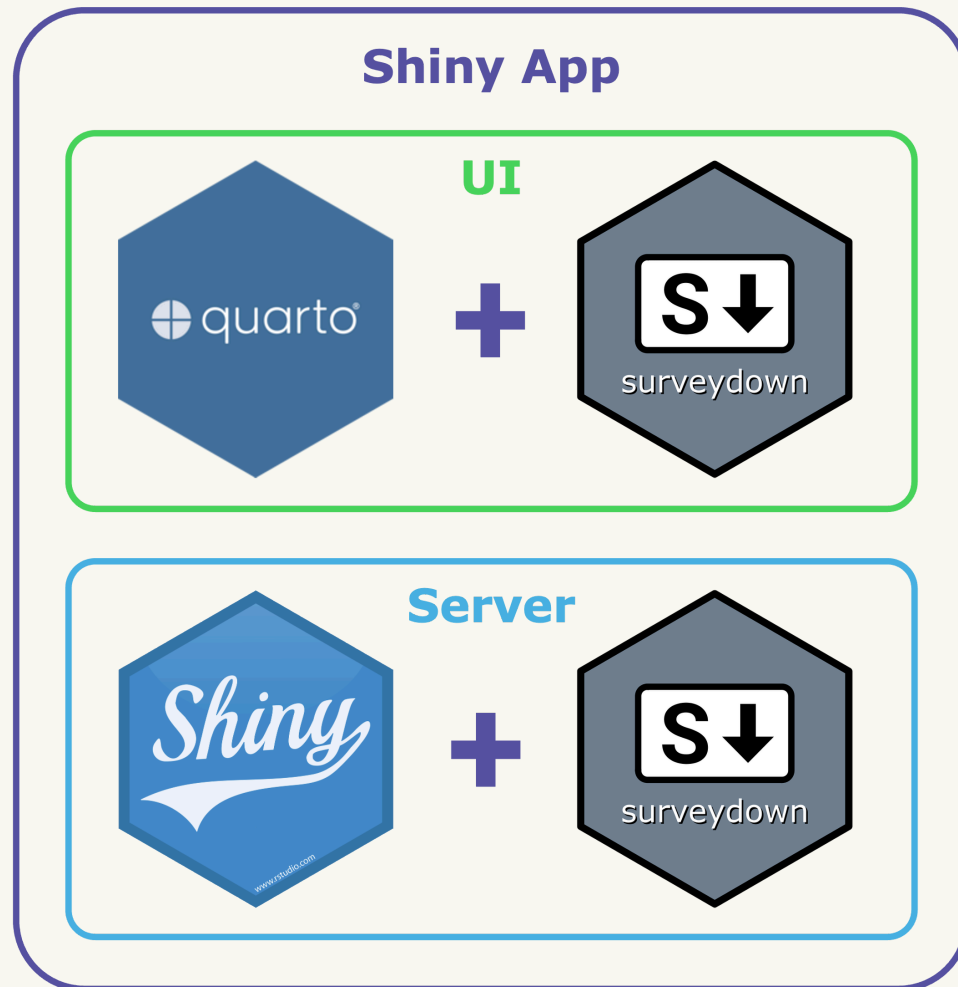


UI



=





A complete surveydown survey

`survey.qmd`

A **Quarto doc** defining the survey content (pages, text, images, questions, etc.).

`app.R`

An **R script** defining the survey Shiny app.

Don't panic!



app.R

```
1 library(surveydown)
2
3 ui ← sd_ui()
4
5 server ← function(input, output, session) {
6
7   sd_server()
8
9 }
10
11 shiny::shinyApp(ui = ui, server = server)
```

app.R

```
1 library(surveydown)
2
3 ui ← sd_ui()
4
5 server ← function(input, output, session) {
6
7   sd_server()
8
9 }
10
11 shiny::shinyApp(ui = ui, server = server)
```

Render `survey.qmd` as UI

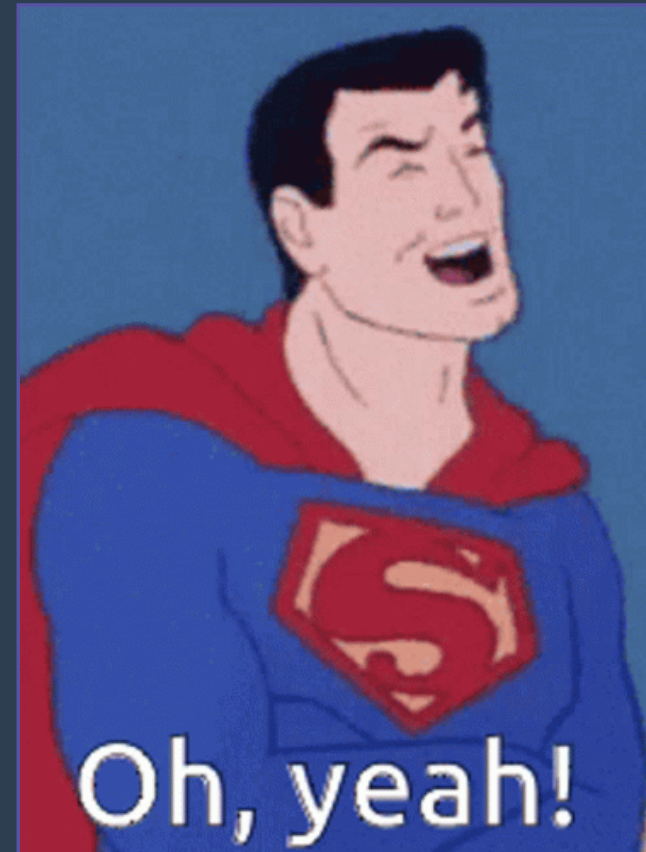
Run the `surveydown` server

Wait a minute...

How do you store the response **data**?



PostgreSQL to the rescue!



supabase.com

The screenshot displays the Supabase Table Editor interface. The left sidebar shows a list of tables in the 'public' schema, with 'superheros' selected. The main area shows the 'superheros' table with columns: session_id, has_fav_hero, fav_hero_name, and hero_universe. The table contains 20 records. The bottom of the interface shows pagination information: Page 1 of 1, 100 rows, 20 records, and buttons for Refresh, Data, and Definition.

session_id	has_fav_hero	fav_hero_name	hero_universe
064585d4b3ce857a4a24e9dea4a657da	no	NULL	NULL
14d82544a547922107f492e6ee9a67dc	no	NULL	NULL
1eb83f38450cc4fa5b3bc8540d161f38	yes	Superman	dc
4ef1e74ae94737c502edb08db8e27f77	yes	NULL	marvel
5cf0f43d847425de2d015e1145ac3199	yes	Spiderman	all
676ae2ebd611f7fe70586c23af5b53c	NULL	NULL	NULL
68fe326c20852efae79bca9b0b089815	no	NULL	NULL
6f436b4a64f154324ad6c4c883e99002	yes	Monkey King	marvel
8352cbbad349838729c489c43f08c4f	yes	Spiderman	dc
86e11fbaa660bc1d94b686fd0206b55d	yes	Superman	all
a426816365182d50dc133ee0c6255b3d	yes	NULL	NULL
a754b629d48ad2c53049c6741e9d4645	no	NULL	NULL
bb7021e8ad7e35be6d75831a2f5c51cd	yes	Spiderman	all
bbe8c56090359bb4eb37e58b95b31316	yes	Spider Man	marvel

app.R

```
1 library(surveydown)
2
3 # sd_db_config()
4 db <- sd_db_connect()
5
6 ui <- sd_ui()
7
8 server <- function(input, output, session) {
9
10   sd_server(db)
11
12 }
13
14 shiny::shinyApp(ui = ui, server = server)
```

Store credentials
Connect to the database

Pass connection to
`sd_server()`

```
demo - main - RStudio

survey.qmd x app.R x analysis.R x
Source on Save Run Source

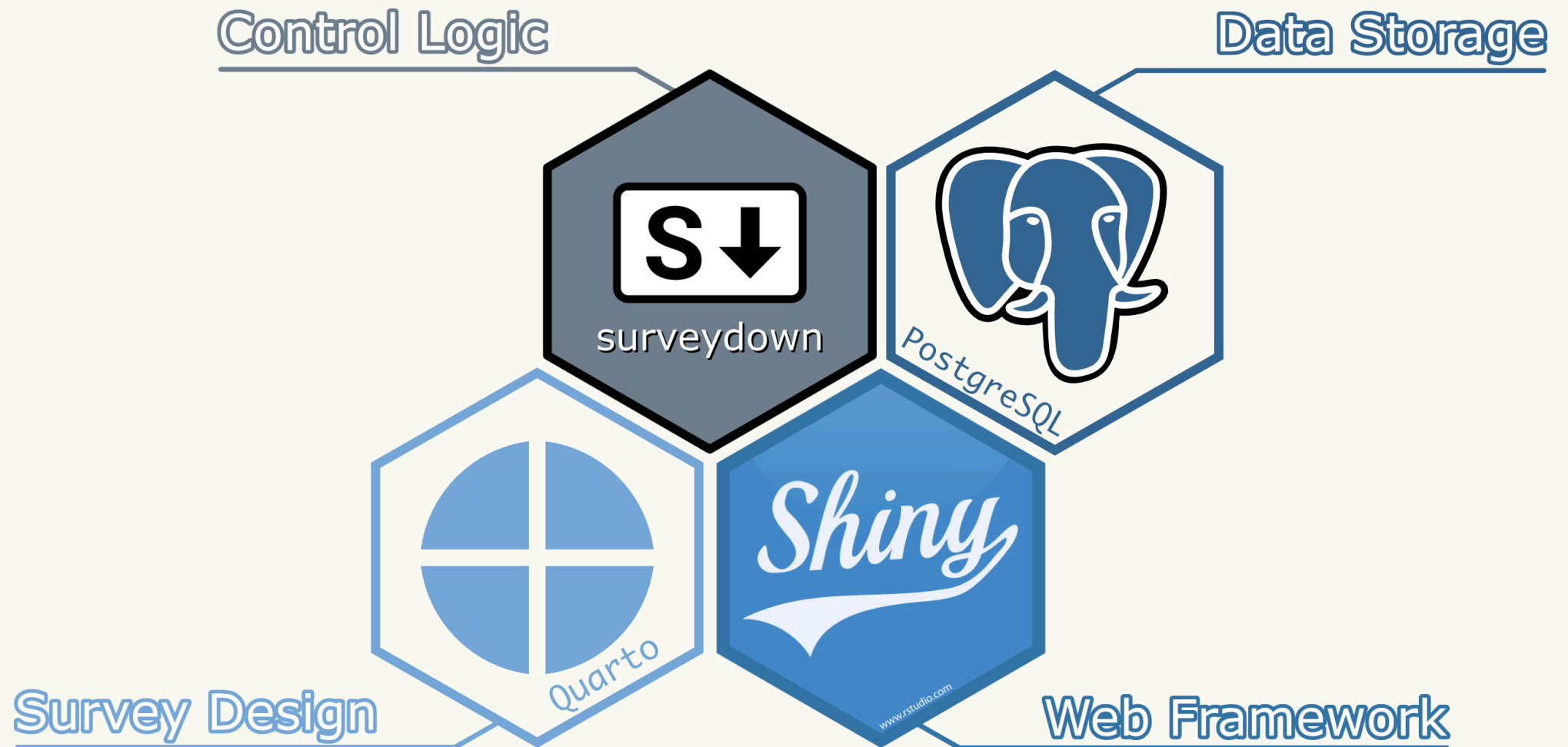
1 library(surveydown)
2 library(dplyr)
3
4 db <- sd_db_connect()
5
6 data <- sd_get_data(db = db)
7
8 glimpse(data)
9

Console Terminal Background Jobs
R 4.4.1 ~ /gh/slides/2025-posit-conf-surveydown/demo/
> db <- sd_db_connect()
✓ Successfully connected to the database.
>
> data <- sd_get_data(db = db)
>
> glimpse(data)
Rows: 20
Columns: 50
$ session_id <chr> "5cf0f43d847425..."
$ time_start <chr> "2025-08-15 15:..."
$ time_end <chr> "2025-08-15 15:..."
$ completion_code <chr> "37081685", "60..."
$ browser <chr> "Safari v26, iO..."
$ ip_address <chr> "140.248.0.0", ...
$ has_fav_hero <chr> "yes", "yes", "..."
$ fav_hero_name <chr> "Spiderman", "S..."
$ hero_universe <chr> "all", "dc", "a..."
$ hero_qualities <chr> "strength, flyi..."
$ dream_power <chr> "telepathy", "t..."
$ responsibility_scale <chr> "10", "8.1", "1..."
$ villain_encounter <chr> "help_backup", ...
$ superhero_rating <chr> NA, NA, NA, NA, ...
$ superhero_rating_spiderman <chr> "love", "love", ...
$ superhero_rating_batman <chr> "love", NA, "li..."
$ superhero_rating_wonder_woman <chr> "love", NA, "li..."

9:1 (Top Level) Copilot: Completion response received. R Script
Environment History Connections Git Tutorial Files Plots Packages Help Viewer Presentation
```

What is **surveydown**?





What can **surveydown** do?

surveydown is feature-packed!

The screenshot shows the surveydown.org website. The browser address bar displays 'surveydown.org'. The navigation bar includes links for 'About', 'Documentation', 'Templates', and 'Blog', along with icons for 'FAQ', a GitHub icon, a RSS icon, a speech bubble icon, and a search icon. A left sidebar contains a 'Documentation' menu with categories like 'Features & Roadmap', 'Survey Basics', 'Question Development', 'Survey Design Concepts', 'Interactivity', and 'Data Management'. The main content area is titled 'Features' and contains a table with four columns: 'Feature', 'Docs', 'Version', and 'Discussion'. The table lists various features such as 'Wide variety of question types', 'Conditionally show questions', 'Markdown formatting for options and buttons', 'Require specific questions or all questions be answered', 'Support for bootstrap themes', 'Customizable scss theme file', 'Ability to Ignore the database connection', 'Time stamps recorded for each question and page interaction', 'Progress bar that updates on each question interaction', 'Customizable progress bar color and position on page', 'Ability to use latest survey results in the survey itself', and 'Pass parameters through the url'. To the right of the table, a section titled 'On this page' lists links for 'surveydown package', 'Features', 'To Do', 'sdstudio package', 'Features', and 'To Do'. The footer of the website states 'Site made with ♥, ♦, and 🍷 quarto'.

Feature	Docs	Version	Discussion
Wide variety of question types	Question Types	v0.0.1	105 , 109
Conditionally show questions	Conditional Display	v0.0.1	
Markdown formatting for options and buttons	Markdown Formatting	v0.0.1	
Require specific questions or all questions be answered	Required Questions	v0.0.2	
Support for bootstrap themes	Themes	v0.0.4	
Customizable scss theme file	Themes	v0.0.4	
Ability to Ignore the database connection	Supabase Ignore	v0.0.8	
Time stamps recorded for each question and page interaction		v0.0.9	
Progress bar that updates on each question interaction	Progress Bar	v0.0.9	
Customizable progress bar color and position on page	Progress Bar	v0.0.9	
Ability to use latest survey results in the survey itself	Fetching Data	v0.1.1	
Pass parameters through the url	Reactive	v0.2.2	92

Feature Highlights

Question types

Conditional logic

12 Built-in Question Types

text

textarea

numeric

mc

mc_multiple

mc_buttons

mc_multiple_buttons

select

slider

slider_numeric

date

daterange

Question Type: text

```
1 sd_question(  
2   type  = "text",  
3   id    = "fav_hero_name",  
4   label = "Who is your favorite superhero?"  
5 )
```



Who is your favorite super hero?

Question Type: mc_buttons

```
1 sd_question(  
2   type = "mc_buttons",  
3   id   = "dream_power",  
4   label = "If you could have ONE superpower, which  
5   option = c(  
6     "🕸 Web-slinging" = "webslinging",  
7     "🛡 Super Strength" = "strength",  
8     "✈ Flight" = "flight",  
9     "🧠 Telepathy" = "telepathy",  
10    "⚡ Super Speed" = "speed"  
11  ),  
12  direction = "vertical"  
13 )
```



If you could have ONE superpower?

 Web-slinging Swing through cities
 Super Strength Lift anything
 Flight Soar through the skies
 Telepathy Read minds
 Super Speed Faster than lightning

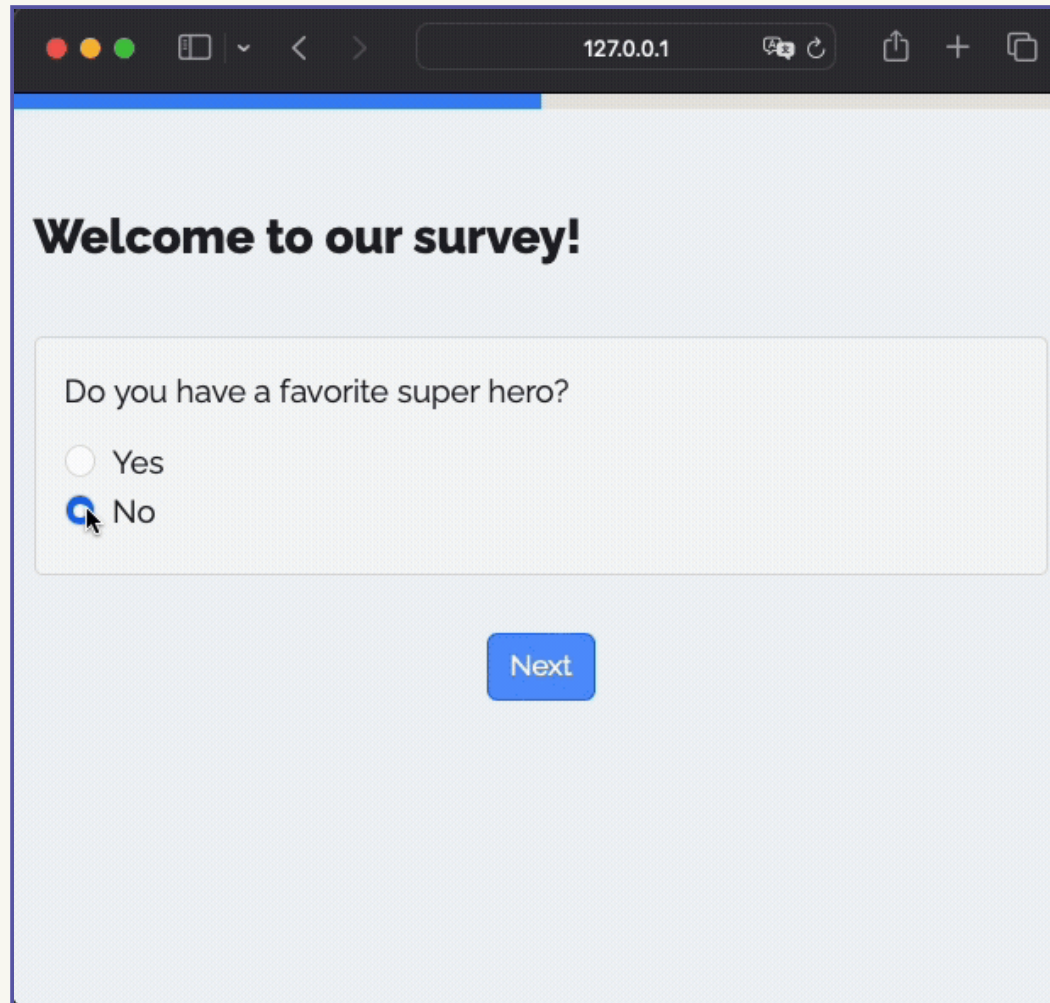
Conditional logic

Conditional showing

Conditional skipping

Conditional stopping

Conditional showing - `sd_show_if()`



A screenshot of a web browser window displaying a survey. The browser's address bar shows the URL `127.0.0.1`. The survey content is as follows:

Welcome to our survey!

Do you have a favorite super hero?

☐ Yes

☒ No

Next

Conditional showing - `sd_show_if()`

survey.qmd

```
1 # Conditional Question
2 sd_question(
3   type = "mc",
4   id    = "has_fav_hero",
5   label = "Do you have a favorite hero?",
6   option = c("Yes" = "yes", "No" = "no")
7 )
8
9 # Target Question
10 sd_question(
11   type = "text",
12   id    = "fav_hero",
13   label = "Who is your favorite superhero?"
14 )
```

app.R

```
1 # Inside server
2 sd_show_if(
3   input$has_fav_hero == "yes" ~ "fav_hero"
4 )
```

Condition ~ Target

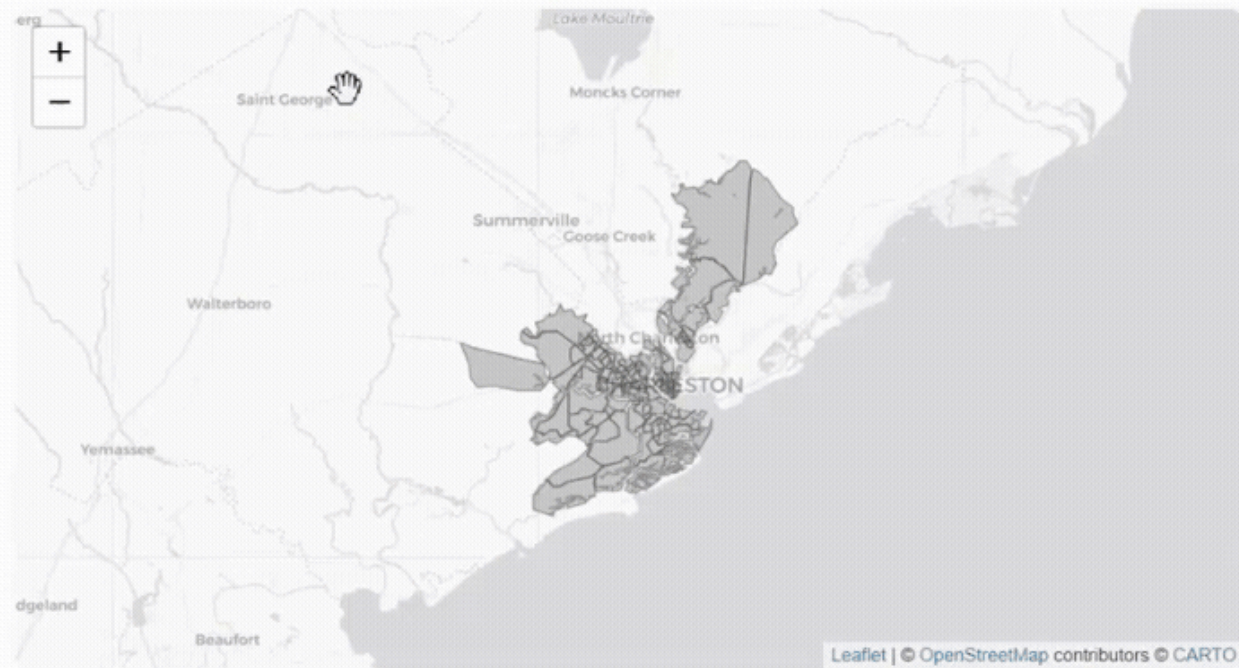
“If {condition}, then show {target}”

If it works in **Shiny**,
it works in **surveydown**.

Embedding an interactive map with **leaflet**:

Question 1: Social Vulnerability is how likely people are to face greater harm or impact on their lives after disruptive hazard events like flooding based on social or economic conditions like income or transportation access.

Please review the map of the City of Charleston, South Carolina, and select the areas you believe are more socially vulnerable to the impacts of hazardous events.

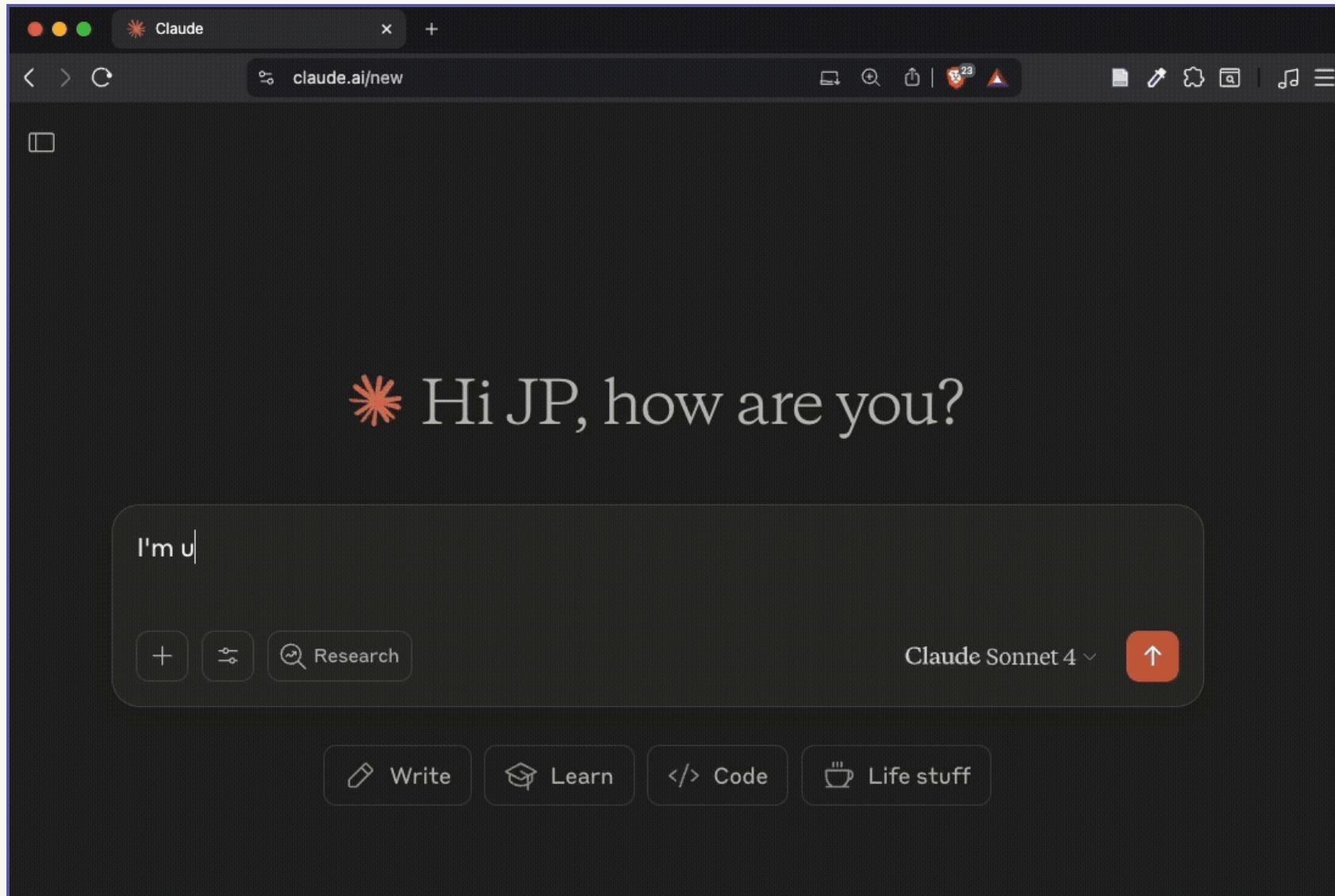


Next Question



surveydown + LLMs
is pretty cool

Generate surveys with LLMs!



What's next?

Try out the companion `sdstudio` package:

```
sdstudio::launch()
```


127.0.0.1
🔍 ↻

surveydown Studio
Build
Preview
Responses

🏠
☁️

🔄
Structure
🔄

⋮
Page: welcome
👤
+T
+Q
|
🔗
🗑️
▼

T
🦸 Welcome to the...
🔗
🗑️

Q
has_fav_hero | mc
🔗
🗑️

Do you have a favorite super hero?

Q
fav_hero_name | text
🔗
🗑️

Who is your favorite super hero?

⋮
Page: hero_details
+T
+Q
|
🔗
🗑️
▼

T
Tell us about your...
🔗
🗑️

Q
hero_universe | mc
🔗
🗑️

Which superhero universe do you prefer?

Q
hero_qualities | mc_multiple
🔗
🗑️

What qualities do you find most appealing in superheroes? (Select all that apply)

⋮
Page: powers
+T
+Q
|
🔗
🗑️
▼

T
If you could have...
🔗
🗑️

Code

survey.qmd
app.R

```

1 ---
2 format: html
3 echo: false
4 warning: false
5 theme: [superhero, custom.scss]
6 barcolor: "#e74c3c"
7 ---
8
9 ```{r}
10 library(surveydown)
11 ```
12
13 ::: {.sd_page id=welcome}
14
15 # 🦸 Welcome to the Ultimate Superhero Survey! 🦸♀️
16
17 Thank you for participating in our research on superhero preferences and
18 powers! This survey will help us understand what makes superheroes
19 appealing to different people.
20
21 **This survey should take about 3-5 minutes to complete.**
22
23 ```{r}
24 sd_question(
25   type = "mc",
26   id   = "has_fav_hero",
27   label = "Do you have a favorite super hero?",
28   option = c("Yes" = "yes", "No" = "no")
29 )

```

surveydown Studio

Build

Preview

Responses

127.0.0.1



Welcome to the Ultimate Superhero Survey!



Thank you for participating in our research on superhero preferences and powers! This survey will help us understand what makes superheroes appealing to different people.

This survey should take about 3-5 minutes to complete.

Do you have a favorite super hero?

☒

Yes

☐

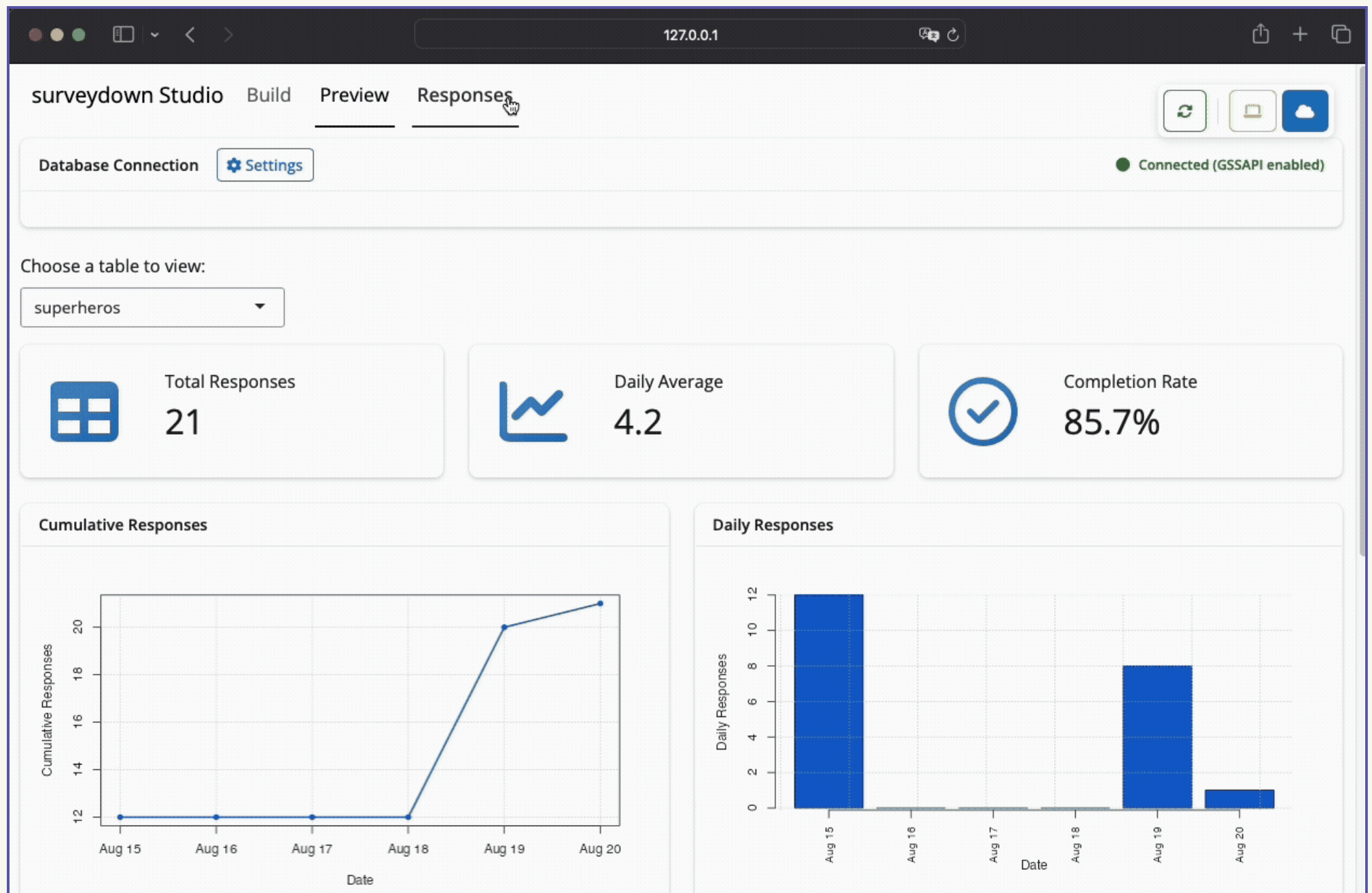
No

Who is your favorite super hero?

Next

Pingfan Hu - Quant UX Con 2025

58



**With great power comes great
responsibility.**

Ways to help:

github.com/surveydown-dev/surveydown/

- Try it out!
- Give it a 🌟
- Post an issue
- Join the GitHub discussion
- Contribute a template



Meet the team!

surveydown.org



John Helveston, Ph.D.



Pingfan Hu



Bogdan Bunea



One more thing...

Workshop by Dr. John Helveston Tomorrow

jhelvy.github.io/2025-qux-surveydown-workshop/

Thursday Nov 6, 2:00 PM UTC – 5, Eastern Time (U.S.)